

# Final Research Blueprint

Consolidated working blueprint for the paper: question, contribution, design, data, experiments, inference, journals, checklist and roadmap

## Machine Learning for Alpha Signal Combination and Portfolio Construction

This report is part of a pre-empirical research proposal. It distinguishes verified evidence, research inference, proposed design choices and hypotheses. No backtest results are reported; expected outcomes are hypotheses.

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# Part I. Blueprint Overview

## 1. Executive Summary

**Topic.** Machine learning for alpha signal combination and portfolio construction.

**Bottom line.** The literature has already shown that ML predicts cross-sectional returns better than linear models (Gu, Kelly & Xiu, 2020). It has shown that ML portfolios are fragile to economic restrictions and trading costs (Avramov, Cheng & Metzker, 2023; Blitz et al., 2023), and that learning portfolio weights under a cost-aware objective substantially improves net-of-cost outcomes (Jensen et al., 2026; DeMiguel et al., 2020; Simon, Weibels & Zimmermann, 2026). Another model will not be a defensible contribution.

**Selected research direction (proposed).** *"Where does machine-learning alpha come from after trading costs?"* A controlled  $2 \times 2 \times 2 \times 2$  factorial design switches four ingredients on and off: nonlinearity, state-dependent signal weights, a cost-aware economic training objective, and forecast-uncertainty shrinkage. Signal library, universe, optimiser, cost model and inference stay fixed. Net-of-cost value is attributed to each ingredient and their interactions using data-snooping-robust tests. Mechanism tests check whether the learned state dependence matches economic theory (Daniel & Moskowitz, 2016; Stambaugh, Yu & Yuan, 2012; Collin-Dufresne, Daniel & Saglam, 2020).

**Why this is the right bet.** The contribution does not depend on beating a benchmark. It survives the closest prior art by nesting those papers as benchmark cells. It uses university-obtainable data. It answers *why*, not only *whether*.

**Primary targets.** JFQA and Management Science (Finance). **Stretch:** RFS. **Alternatives:** RAPS, Review of Finance, Quantitative Finance. **Backups:** JFM, JBF, JEF.

**Timeline.** About 16 months to a submitted working paper, with a 2021-2025 lockbox holdout opened once.

## 2. State of the Literature

Five waves (detail in PDF 1):

1. The factor zoo and multiple testing (Cochrane, 2011; Harvey, Liu & Zhu, 2016; Hou, Xue & Zhang, 2020; Jensen, Kelly & Pedersen, 2023; Chen & Zimmermann, 2022; McLean & Pontiff, 2016).
2. Linear aggregation and shrinkage (Lewellen, 2015; Light, Maslov & Rytchkov, 2017; Kozak, Nagel & Santosh, 2020; Freyberger, Neuhierl & Weber, 2020).
3. The ML prediction wave (Gu, Kelly & Xiu, 2020; Gu, Kelly & Xiu, 2021; Kelly, Pruitt & Su, 2019; Chen, Pelger & Zhu, 2024; Bryzgalova, Pelger & Zhu, 2025; Kelly, Malamud & Zhou, 2024).
4. The implementability reckoning (Novy-Marx & Velikov, 2016; Avramov, Cheng & Metzker, 2023; Blitz et al., 2023; Chen & Velikov, 2023; Azevedo, Hoegner & Velikov, 2023; Chen, Hanauer & Kalsbach, 2024; Chen & Welch, 2026).
5. Learning under the economic objective (Brandt, Santa-Clara & Valkanov, 2009; DeMiguel et al., 2020; Simon, Weibels & Zimmermann, 2026; Jensen et al., 2026; Butler & Kwon, 2023; Cong et al., 2021).

Parallel strands: factor timing and regimes (Haddad, Kozak & Santosh, 2020; Neuhierl et al., 2024; DeMiguel, Martin-Utrera & Uppal, 2024; Collin-Dufresne, Daniel & Saglam, 2020), uncertainty (Barroso & Saxena, 2022; Liao et al., 2025), and non-stationarity (Capponi et al., 2025; Wang & Lera, 2025).

## 3. What Has Already Been Solved

- ML beats linear models on prediction and gross spreads, via nonlinear interactions.
- Trading costs and turnover are first-order. Horizon choice and banding help.
- Cost-aware learning of weights beats predict-then-optimize net of costs.
- Factor returns are timeable at the factor level. A conditional multifactor portfolio survives costs.
- Tools for backtest integrity exist: DSR, PBO, SPA, MCS, FDR.

## 4. What Remains Unsolved

- Which ingredient of an ML pipeline creates net value, and how the ingredients interact [inference; re-verify before submission].
- Whether state dependence helps only when costs are internalised.
- Whether learned state dependence is economically interpretable.
- How forecast uncertainty interacts with costs.

## 5. Most Important Research Gaps

| Gap | Statement                                                                                           | Closest prior art                   |
|-----|-----------------------------------------------------------------------------------------------------|-------------------------------------|
| G1  | No controlled attribution of net ML value across nonlinearity, conditioning, objective, uncertainty | JKMP2026; SWZ2023; GKX2020; CHK2024 |
| G2  | Conditioning × cost-internalisation interaction untested at stock level                             | DMU2024; CDS2020                    |
| G3  | Economic interpretation of learned state dependence                                                 | DM2016; SY2012; CPZ2024             |
| G4  | Uncertainty shrinkage and turnover channel under costs                                              | LMNS2025; BS2022                    |
| G5  | Stock-level implementation of factor-timing information                                             | NRRZ2024; HKS2020                   |

## 6. Fifteen Possible Project Ideas (ranked by weighted score)

| Rank | ID  | Proposed title                                                                                                                                | Nov. | Acad. | Prac. | Pub. | Data | Impl. diff. | Risk | Over all | Target journals                                                                           |
|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|------|-------|-------|------|------|-------------|------|----------|-------------------------------------------------------------------------------------------|
| 1    | I01 | Where Does Machine-Learning Alpha Come From After Trading Costs? Nonlinearity, State Dependence and Cost-Aware Learning in Signal Combination | 8    | 9     | 9     | 8    | 8    | 7           | 4    | 7.55     | RFS (stretch); JFQA, Management Science (target); Quantitative Finance, JFM, JBF (backup) |
| 2    | I02 | Forecasting Signal Efficacy                                                                                                                   | 7    | 7     | 9     | 7    | 8    | 6           | 6    | 6.75     | RAPS, JFQA (target); JBF, JEF, IJF (backup)                                               |
| 3    | I07 | Netting-Aware Signal Selection                                                                                                                | 6    | 6     | 9     | 6    | 9    | 5           | 4    | 6.65     | JFQA, FAJ, JFM                                                                            |
| 4    | I03 | Uncertainty-Aware Alpha Combination                                                                                                           | 6    | 6     | 8     | 6    | 8    | 6           | 5    | 6.20     | Quantitative Finance, JFEc (target); JEF, EJOR (backup)                                   |
| 5    | I13 | State-Dependent Capacity of ML Alpha                                                                                                          | 6    | 6     | 9     | 6    | 7    | 6           | 5    | 6.15     | JFM, FAJ                                                                                  |
| 6    | I15 | Alpha-Risk Model Misalignment in ML Portfolios                                                                                                | 6    | 5     | 8     | 5    | 8    | 5           | 5    | 5.95     | QF, JPM, FAJ                                                                              |
| 7    | I14 | Publication-Aware Training: Eliminating Look-Ahead in Signal Discovery                                                                        | 4    | 5     | 7     | 5    | 9    | 4           | 3    | 5.90     | CFR, JFM                                                                                  |
| 8    | I04 | Differentiable Cost-Aware Portfolio Layers for High-Dimensional Equity Signal Combination                                                     | 7    | 6     | 7     | 6    | 8    | 9           | 7    | 5.80     | Management Science, Operations Research, INFORMS JOC (target); QF, EJOR (backup)          |
| 9    | I05 | Regime-Gated Mixture-of-Experts for Alpha Signal Combination                                                                                  | 6    | 5     | 8     | 6    | 8    | 7           | 7    | 5.75     | QF, JBF, JEF (target)                                                                     |
| 10   | I06 | Learning Signal Decay: Horizon-Aware Combination with Aim Portfolios                                                                          | 5    | 5     | 8     | 5    | 8    | 6           | 5    | 5.65     | QF, JFM                                                                                   |
| 11   | I12 | Bayesian Dynamic Model Averaging of Alpha Signals                                                                                             | 4    | 4     | 7     | 5    | 9    | 4           | 4    | 5.65     | IJF, JEF                                                                                  |
| 12   | I09 | Transfer Learning of Alpha Combination Across 90+ Markets                                                                                     | 6    | 6     | 7     | 6    | 6    | 7           | 6    | 5.60     | JFQA, RF, JBF                                                                             |
| 13   | I10 | Do Text and LLM Signals Add Net-of-Cost Value to a Mature Signal Library?                                                                     | 7    | 6     | 8     | 6    | 4    | 7           | 7    | 5.50     | JFQA, MS                                                                                  |
| 14   | I08 | Crowding-Aware Alpha Combination                                                                                                              | 6    | 5     | 8     | 5    | 6    | 6           | 7    | 5.35     | JFM, JBF                                                                                  |
| 15   | I11 | Graph Neural Signal Propagation with Trading Costs                                                                                            | 6    | 5     | 6     | 5    | 4    | 8           | 7    | 4.65     | QF, EJOR                                                                                  |

## 7. Top Five Paper Concepts

| Concept | Title                                               | Core question                                                             | Target journals         | Difficulty  |
|---------|-----------------------------------------------------|---------------------------------------------------------------------------|-------------------------|-------------|
| A       | Where Does ML Alpha Come From After Trading Costs?  | Attribution of net value to four ingredients                              | RFS (stretch); JFQA, MS | High        |
| B       | Forecasting Signal Efficacy                         | Can signal-level meta-features time signals at stock level after netting? | RAPS, JFQA; JBF, JEF    | Medium-high |
| C       | Uncertainty-Aware Alpha Combination                 | Does forecast uncertainty improve net portfolios via turnover?            | QF, JFEc                | Medium      |
| D       | Differentiable Cost-Aware Portfolio Layers at Scale | Does end-to-end training scale and help net?                              | MS, OR, EJOR            | High        |
| E       | Regime-Gated Mixture-of-Experts                     | Do interpretable state gates add net value?                               | QF, JBF, JEF            | Medium      |

## 8. Best Final Research Idea

**Concept A.** It scored highest on novelty, academic contribution, rigor, reproducibility, mechanism and journal fit (mean 8.2/10 vs 7.5 for the runner-up). It nests Concepts B, C and E. Its closest competitors become benchmark cells, and its value holds whatever the empirical outcome.

### Contribution statement (draft, not overclaiming).

*Existing literature shows ML improves return prediction and that cost-aware learning improves the implementable frontier. However, it compares bundled pipelines on differing data, costs and criteria. We propose a factorial attribution that isolates nonlinearity, state dependence, the economic objective and uncertainty within one controlled environment. This lets us test whether state dependence helps only when costs are internalised, and whether uncertainty works through turnover. We will report evidence across US equities 1995-2025 (with a 2021-2025 lockbox), three AUM levels, cost perturbations and developed international markets. Implications will be stated only after the lockbox is opened.*

## 9. Exact Methodology (summary; full specification in Part II)

- **Universe.** CRSP common stocks (share codes 10/11) on NYSE/AMEX/NASDAQ; main filter excludes price < \$5 and stocks below the NYSE 20th size percentile.
- **Frequency.** Monthly signals and rebalancing; daily data for risk and costs.
- **Periods.** Training from 1972 (expanding); 5-year rolling validation; walk-forward test 1995-2020; lockbox 2021-2025; annual refit; embargo equal to the horizon.
- **Signals.** About 150 JKP characteristics in 13 themes, rank-normalised, with missing indicators and publication flags.
- **Cells.** {Linear, Nonlinear} × {Static, State-conditional} × {Prediction loss, Economic net-of-cost loss} × {No uncertainty, Ensemble shrinkage}.
- **Models.** Ridge; LightGBM + NN3 ensembles; gated mixture-of-experts; linear and NN parametric policies with costs.
- **Portfolio.** Mean-variance minus costs; dollar/beta/industry neutral; position, trade and ADV limits; structural factor risk model.
- **Costs.** Half effective spread +  $k \cdot \sigma \cdot |Q|^{\wedge 1.5} / \text{ADV}^{\wedge 0.5} + 1$  bp; borrow fees; AUM \$0.1bn/\$1bn/\$10bn; k in {0.5, 1, 2, 3}.

## 10. Data Requirements

CRSP (monthly/daily, delistings), Compustat + CCM, JKP characteristics (WRDS) with OpenAP cross-checks, TAQ effective spreads (1993+) and EDGE/Abdi-Ranaldo estimators, Ken French/q/JKP factors, and FRED, Goyal, VIX, Baker-Wurgler and Pastor-Stambaugh state variables. Optional: IBES, 13F, short interest, Markit borrow fees, JKP international. Bias register in Part II and PDF 4.

## 11. Experimental Design

Forty experiments across eight blocks (E00-E64). Data integrity -> signal library -> baselines -> 16-cell factorial core with decomposition -> portfolio and costs -> inference -> robustness -> interpretation. Every configuration is logged. A specification freeze precedes a single lockbox evaluation.

## 12. Benchmark Framework

Naive composites (EW, theme-EW, IC/ICIR-weighted, inverse-vol); linear (Lewellen FM, ridge, ENet, PLS); portfolio-level (KNS SDF; NRRZ-style timing); nonlinear ML (RF, LightGBM, GKX NN3, IPCA); economic objective (DMNU linear PPP, JKMP Portfolio-ML from authors' code, deep PPP); cost mitigation (NMV banding); 1/N and market; factor-model alphas (FF5+UMD, q5, JKP themes). All benchmarks get equal tuning budgets and the same costs.

## 13. Statistical Framework

Newey-West t-stats; Ledoit-Wolf Sharpe-difference bootstrap; Clark-West and Diebold-Mariano; Hansen SPA, White Reality Check, Romano-Wolf StepM, Model Confidence Set; Deflated Sharpe and PBO over the full trial log; Harvey-Liu haircuts; factorial contrasts and Shapley attribution with block-bootstrap CIs; Bai-Perron stability; BH-FDR; non-standard errors across a pre-declared multiverse.

## 14. Robustness Framework

Subperiods; universes (all, ex-micro, top-500); horizons and rebalancing; cost multipliers and spread estimators; AUM grid; risk models; rolling vs expanding windows; publication-gated signals; placebo states; NN seeds and grid perturbations; long-only; international developed markets; alternative uncertainty estimators.

## 15. Journal Strategy

Write for finance reviewers. Present at AFA, WFA, EFA, SFS Cavalcade, NBER, SoFiE and ICAIF before submission. Submit to RFS only if the attribution yields a general lesson backed by mechanisms; otherwise JFQA or MS first; then RAPS or Review of Finance, then QF, JFM, JBF or JEF. Practitioner spin-off in FAJ or JFDS after SSRN posting. Detail in PDF 2.

## 16. Target Journal List

| Role                | Journals                                                                                 |
|---------------------|------------------------------------------------------------------------------------------|
| Stretch             | Review of Financial Studies; Journal of Finance                                          |
| Primary targets     | Journal of Financial and Quantitative Analysis; Management Science (Finance)             |
| Strong alternatives | Review of Asset Pricing Studies; Review of Finance; Quantitative Finance                 |
| Backups             | Journal of Financial Markets; Journal of Banking & Finance; Journal of Empirical Finance |
| Spin-offs           | FAJ / JFDS (practitioner); MS/OR/EJOR (optimiser methods); JFEC (inference)              |

## 17-18. Reviewer-Proof Checklist and Implementation Roadmap

See Parts IV and V of this blueprint.

## 19. Citations

All in-text citations resolve to the reference list at the end of this document. The full bibliography (BibTeX and formatted), with Crossref verification status for every DOI, is in o8\_References.

## 20-21. Generated Reports and Workbooks

| Deliverable | Path                                                    | Contents                                                                                                                |
|-------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| PDF 1       | 09_Reports/PDF1_Comprehensive_Literature_Review.pdf     | Executive summary, literature evolution, key papers, methods, findings, gaps, BlackRock AIM analysis, matrix appendices |
| PDF 2       | 09_Reports/PDF2_Journal_Strategy.pdf                    | Journal tiers, comparison, fit, expectations, ranking, submission strategy, reviewer checklist                          |
| PDF 3       | 09_Reports/PDF3_Research_Gap_and_Opportunity_Report.pdf | Existing approaches, weaknesses, gaps, pipelines A-F, prior-art tracker, 15 ideas, 5 concepts                           |
| PDF 4       | 09_Reports/PDF4_Proposed_Research_Design.pdf            | Research question, hypotheses, architecture, data, ML/portfolio methods, benchmarks, experiments, statistics            |
| PDF 5       | 09_Reports/PDF5_Final_Research_Blueprint.pdf            | Consolidated working blueprint (overview + design + framework + checklist + roadmap)                                    |
| Workbook 1  | 09_Reports/Workbook1_Literature_Database.xlsx           | 200 entries; summary formulas; closest prior art; legend                                                                |
| Workbook 2  | 09_Reports/Workbook2_Journal_Analysis.xlsx              | Journal analysis with priority formulas, role formatting, metrics note                                                  |
| Workbook 3  | 09_Reports/Workbook3_Research_Idea_Matrix.xlsx          | 15 ideas; editable weights; score and rank formulas; concept selection                                                  |
| Workbook 4  | 09_Reports/Workbook4_Experiment_Matrix.xlsx             | Experiment register with status dropdowns, progress summary, factorial contrasts, Gantt, checklist tracker              |

# Part II. Research Design: Where Does Machine-Learning Alpha Come From After Trading Costs?

*Working title: "Nonlinearity, State Dependence, and Cost-Aware Learning: Decomposing the Net Value of Machine-Learning Signal Combination".*

Everything in this document is a **proposed** design. Expected outcomes are **hypotheses**, not results.

## 1. Research Question

How much of the net-of-cost investment value of machine-learning signal combination is attributable to (i) **nonlinear** signal interactions, (ii) **state-dependent** signal weights, (iii) a **cost-aware economic training objective**, and (iv) **forecast-uncertainty** shrinkage? Do these ingredients interact? Is the learned state dependence consistent with economic theory?

## 2. Hypotheses (pre-registered)

| ID | Hypothesis                                                                                                                                                                                          | Primary test                                                                               | Grounding                                                                                                   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| H1 | Nonlinearity raises gross IC and gross Sharpe relative to linear combination, but at least 50% of the gross Sharpe gain disappears net of costs                                                     | Clark-West (IC/R <sup>2</sup> ), LW2008 Sharpe difference gross vs net, factorial contrast | Gu, Kelly & Xiu (2020); Avramov, Cheng & Metzker (2023); Blitz et al. (2023)                                |
| H2 | State-dependent weights add net value only when costs are internalised in training (positive State × Objective interaction)                                                                         | Factorial interaction contrast, block bootstrap; SPA                                       | Collin-Dufresne, Daniel & Saglam (2020); Garleanu & Pedersen (2013); DeMiguel, Martin-Utrera & Uppal (2024) |
| H3 | The cost-aware objective is the largest main effect on net utility                                                                                                                                  | Factorial main effects; Shapley attribution                                                | Jensen et al. (2026); DeMiguel et al. (2020)                                                                |
| H4 | Uncertainty shrinkage raises net Sharpe, mainly by reducing turnover                                                                                                                                | Mediation: turnover change vs gross alpha change                                           | Liao et al. (2025); Barroso & Saxena (2022)                                                                 |
| H5 | Learned state dependence matches theory: lower momentum weight after market declines with high volatility; stronger short-leg anomaly weights after high sentiment; faster trading in liquid states | Regressions of implied weights / turnover on states                                        | Daniel & Moskowitz (2016); Stambaugh, Yu & Yuan (2012); Collin-Dufresne, Daniel & Saglam (2020)             |
| H6 | Gross ML gains load on limits-to-arbitrage stocks; cost-aware learning shifts exposure toward liquid stocks                                                                                         | Characteristic tilts of portfolios                                                         | Avramov, Cheng & Metzker (2023)                                                                             |

**Null results are informative.** If H2 fails, the paper reports that conditioning adds no net value even with costs internalised. That is an important negative finding for practitioners (e.g., for the regime resampling in BlackRock Systematic (2025)).

## 3. Architecture

### Empirical research pipeline (shaded stages = factorial ML core)

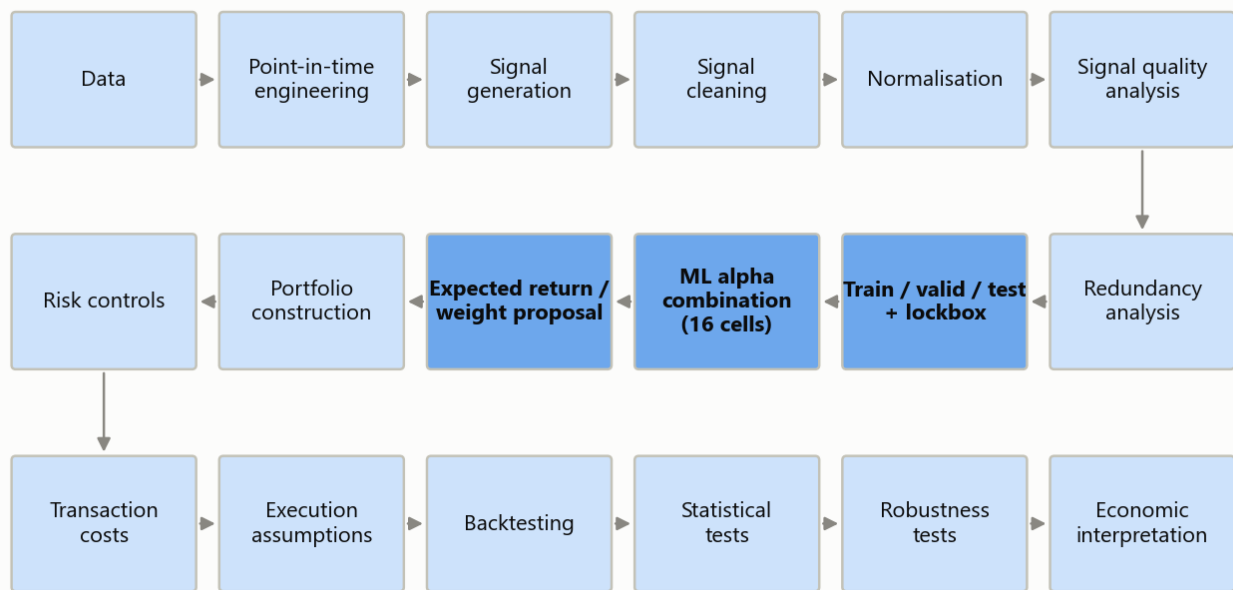


Figure 1. End-to-end empirical pipeline.

|                   |                | Prediction loss<br>no uncertainty | Prediction loss<br>+ uncertainty | Economic loss<br>no uncertainty | Economic loss<br>+ uncertainty |
|-------------------|----------------|-----------------------------------|----------------------------------|---------------------------------|--------------------------------|
| Linear, static    | L-S-P-0<br>E20 |                                   | L-S-P-U<br>E20 + E28             | L-S-E-0<br>E24                  | L-S-E-U<br>E24 + E28           |
|                   | L-C-P-0<br>E22 |                                   | L-C-P-U<br>E22 + E28             | L-C-E-0<br>E26                  | L-C-E-U<br>E26 + E28           |
| Nonlinear, static | N-S-P-0<br>E21 |                                   | N-S-P-U<br>E21 + E28             | N-S-E-0<br>E25                  | N-S-E-U<br>E25 + E28           |
|                   | N-C-P-0<br>E23 |                                   | N-C-P-U<br>E23 + E28             | N-C-E-0<br>E27                  | N-C-E-U<br>E27 + E28           |

Economic loss = net-of-cost utility. All 16 cells share signals, universe, splits, optimiser, risk and cost model.  
Dark cell = full model; E29 decomposes main effects and interactions.

Figure 2. The 16-cell factorial design (experiment IDs from Workbook 4).

Point-in-time data (CRSP, Compustat, JKP/OpenAP signals, state variables, cost inputs)

- > Signal construction (~150 signals, 13 themes) + publication-date flags
- > Cross-sectional rank normalisation, missing-value indicators
- > Signal quality & redundancy diagnostics (IC, decay, turnover, clusters)
- > Walk-forward splits (expanding train, 5y validation, embargo = horizon, annual refit; 2021-2025 lockbox)
- > 16-cell factorial combination models
  - Form: Linear (L) | Nonlinear (N)
  - Conditioning: Static (S) | State-conditional (C)
  - Objective: Prediction loss (P) | Economic net-of-cost loss (E)
  - Uncertainty: None (0) | Ensemble-uncertainty shrinkage (U)
- > P-cells: standardized scores -> common alpha scaling -> common TC-aware optimiser
- > E-cells: weights learned under net utility, then projected onto the same constraint set
- > Risk controls (neutralities, position/trade limits, factor risk model)
- > Execution & costs (spread + power-law impact; AUM grid; borrow costs)
- > Backtest -> statistics (SPA/StepM/MCS, LW2008, DSR, PBO, NW alphas)
- > Factorial & Shapley attribution -> mechanism tests -> robustness & multiverse

**Design principle (proposed).** All cells share the signal library, universe, splits, constraint set, risk model, cost model, AUM and evaluation code. The only differences between cells are the four switched ingredients, so differences in net performance can be attributed to them.

## 4. Data

| Component                                   | Source                                                                                                                                                                                          | Access                                  | Notes                                                                                |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------|
| Returns, prices, volume, shares, delistings | CRSP US Stock (monthly & daily) via WRDS                                                                                                                                                        | University subscription                 | Share codes 10/11; exchanges NYSE/AMEX/NASDAQ; delisting returns applied             |
| Fundamentals                                | Compustat North America (annual & quarterly) via WRDS                                                                                                                                           | Subscription                            | Lagged availability (see Section 9); CCM link table                                  |
| Characteristics                             | JKP global characteristics (WRDS; code public) and/or Open Source AP firm-level signals (209 downloadable + 3 from CRSP)                                                                        | WRDS / free download                    | JKP factor returns under CC BY-NC 4.0 (jkpfactors.com); OpenAP data through Dec 2024 |
| Factor benchmarks                           | Ken French data library; q-factors (global-q.org); JKP theme factors                                                                                                                            | Free                                    | FF5+UMD, q5, JKP themes                                                              |
| State variables                             | FRED (credit/term spreads, CPI, yields), Goyal predictor data, CBOE VIX (from 1990), Baker-Wurgler sentiment, Pastor-Stambaugh liquidity                                                        | Free (verify latest update dates)       | All lagged one month                                                                 |
| Trading-cost inputs                         | Chen-Velikov effective spreads (hf-spreads-all code, TAQ-based); EDGE OHLC estimator (Ardia, Guidotti & Kroencke, 2024); Abdi-Rinaldo (Abdi & Rinaldo, 2017); Hasbrouck Gibbs (Hasbrouck, 2009) | TAQ via WRDS; estimators from CRSP OHLC | Hierarchy: TAQ effective spread (1993+), else EDGE, else AR                          |
| Borrow costs (optional)                     | Markit/IHS securities finance via WRDS if licensed (JKMP code references short-selling fees from WRDS)                                                                                          | Institution-dependent                   | Fallback: institutional-ownership proxy with fee tiers                               |

## 5. ML Methodology

**Inputs.** For stock  $i$  at month-end  $t$ : signal vector  $x_{\{i,t\}}$  (about 150 cross-sectionally rank-normalised to  $[-0.5, 0.5]$ ; missing set to 0 with 13 theme-level missing indicators); cost features  $c_{\{i,t\}}$  (log effective spread, log dollar volume, Amihud illiquidity, 60-day volatility); state vector  $z_t$  (pre-specified, Section 6).

**Targets.** Prediction cells: cross-sectionally demeaned excess return over horizon  $h$  in  $\{1, 3, 6, 12\}$  months ( $h$  chosen in validation and counted as a trial), rank-transformed in robustness. Economic cells: realised next-month net portfolio utility.

### Model classes per cell.

- **L-S-P.** Ridge on  $x$  ( $\lambda$  grid  $10^{-4}$  to  $10^4$ , 17 points). Elastic net and PLS reported as baselines.
- **L-C-P.** Ridge on  $[x, xxz]$  with separate penalties for main and interaction blocks.
- **N-S-P.** LightGBM (num\_leaves  $\{15, 31, 63\}$ ; learning\_rate  $\{0.01, 0.05\}$ ; min\_data\_in\_leaf  $\{500, 2000\}$ ; feature\_fraction  $\{0.5, 0.8\}$ ; bagging 0.8; L2  $\{0, 10\}$ ; early stopping on validation, up to 2,000 trees) and an NN3 ensemble (32-16-8 ReLU, batch norm, Adam lr  $1e-3$ , L1  $\{1e-5, 1e-4, 1e-3\}$ , early stopping patience 5, 10 seeds). The pre-specified nonlinear prediction is the equal-weight average of the LightGBM and NN ensembles; each is also reported separately.
- **N-C-P.** As N-S-P with  $z$  and  $xxz$  appended for NN and  $z$  appended for LightGBM. Gated mixture-of-experts variant: 3 NN experts, softmax gate on  $z$  only (interpretable gate).
- **L-S-E.** Linear parametric portfolio policy with costs (DeMiguel et al., 2020).
- **L-C-E.** Linear policy with coefficients linear in  $z$  (conditional PPP).
- **N-S-E.** NN policy (NN3) mapping  $(x, c)$  to weight tilts, trained on net utility (Simon, Weibels & Zimmermann, 2026; Jensen et al., 2026 Portfolio-ML as external benchmark).
- **N-C-E.** NN policy on  $(x, c, z)$  with state gate.

- **Uncertainty (U).** For P-cells, the alpha score is shrunk as  $\alpha_{\text{shrunk}} = \alpha / (1 + \kappa \cdot s^2 / s_{\text{avg}}^2)$ , where  $s$  is the stock's ensemble standard deviation across 10 members,  $s_{\text{avg}}$  is its cross-sectional average (seeds for NN, bagged subsamples for LightGBM or ridge) and  $\kappa$  in  $\{0, 0.5, 1, 2, 4\}$  is chosen in validation. For E-cells, a penalty proportional to the ensemble variance of proposed weights is added. A bootstrap-SE variant (Liao et al., 2025) is used in robustness.

**Economic loss (E-cells).** Over training months, maximise the mean of

$$U_t = w_t' r_{t+1} - (\gamma/2) \cdot w_t' \Sigma_t w_t - TC_t(w_t - w_{+t-1}) - \eta \cdot \|w_t\|_1$$

where  $w_{+t-1}$  are drifted previous weights.  $TC$  is the cost model of Section 7 with a smooth (Huber) approximation of  $|\cdot|$  for training.  $\gamma$  is calibrated so the ex-ante volatility of the validation-period portfolio is 10% annualised. Weights are normalised to gross exposure 2 (long 1, short 1) and projected onto the Section 7 constraint set before evaluation. Training uses truncated back-propagation through 12-month sequences so the turnover term is learned.

**Alpha scaling for P-cells (common across models).** score  $\rightarrow$  z-score cross-sectionally  $\rightarrow \alpha_{\{i,t\}} = IC_v \cdot \sigma_{\{i,t\}} \cdot \text{score}_{\{i,t\}}$  (Grinold-Kahn convention), where  $IC_v$  is the model's validation-period IC. This prevents a model from "winning" by producing larger forecast magnitudes.

## 6. State Variables (fixed ex ante; all observable at $t$ )

| Variable | Definition                                                                     | Economic rationale                                                                                                   |
|----------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| MKTVOL   | log 21-day realised volatility of CRSP VW index                                | Momentum crashes; risk prices fall with volatility (Daniel & Moskowitz, 2016; DeMiguel, Martin-Utrera & Uppal, 2024) |
| BEAR     | 1 if CRSP VW cumulative 24-month return < 0                                    | Momentum crash state (Daniel & Moskowitz, 2016)                                                                      |
| ILLIQ    | VW average Amihud illiquidity, detrended (log deviation from 12-month average) | Liquidity regimes (Collin-Dufresne, Daniel & Saglam, 2020; Amihud, 2002)                                             |
| SENT     | Baker-Wurgler sentiment (orthogonalised version), last available               | Short-leg anomaly strength (Stambaugh, Yu & Yuan, 2012; Baker & Wurgler, 2006)                                       |
| CREDIT   | BAA – AAA yield spread                                                         | Distress / funding                                                                                                   |
| TERM     | 10y – 3m Treasury                                                              | Business cycle                                                                                                       |
| INFL     | CPI YoY inflation (lagged for release)                                         | Inflation regime                                                                                                     |
| DRATE    | 12-month change in 10y yield                                                   | Rate regime                                                                                                          |
| FMOM_k   | Trailing 12-month return of theme-k factor (13 values)                         | Factor momentum (Ehsani & Linnainmaa, 2022)                                                                          |
| DISP     | Cross-sectional dispersion of composite signal                                 | Opportunity set                                                                                                      |

State variables are standardised with expanding-window means and standard deviations. HMM-smoothed regime probabilities are prohibited (look-ahead). Placebo tests replace  $z$  with (i) block-shuffled  $z$  and (ii) random-walk series.

## 7. Portfolio Methodology

**Optimiser (P-cells and projection step for E-cells).**

$$\max_w \alpha_t' w - (\gamma/2) \cdot w' \Sigma_t w - \sum_i TC_i(|w_i - w_{+i,t-1}|)$$

subject to:

- $\sum_i w_i = 0$  (dollar neutral) and  $\sum_i |w_i| \leq 2$
- $|\beta_t' w| \leq 0.05$  (beta from the risk model)
- $|\sum_{i \in g} w_i| \leq 0.02$  for each Fama-French 49 industry  $g$
- $|w_i| \leq \min(1\%, 10 \times \text{participation cap})$ , with trade size  $|\Delta w_i| \cdot \text{AUM} \leq 5\%$  of 20-day  $\text{ADV}_i$
- Long-only benchmark-relative variant in robustness:  $w \geq 0$ ,  $\sum w = 1$ , tracking error  $\leq 4\%$

Solved with CVXPY (MOSEK or Clarabel); failures are logged and the portfolio is held.

**Risk model.** Structural factor model  $\Sigma = B F B' + D$ .  $B$  is market beta, 13 JKP theme exposures (cross-sectional z-scores) and FF49 industry dummies. Factor returns come from daily cross-sectional WLS ( $\sqrt{\text{market-cap}}$  weights).  $F$  is EWMA with a 126-day half-life and Newey-West adjustment.  $D$  is EWMA idiosyncratic variance with Bayesian shrinkage toward size-decile means. Robustness: a 20-factor PCA statistical model and a Ledoit-Wolf-shrunk sample covariance on the top-500 universe (Ledoit & Wolf, 2017).

#### Transaction-cost model (per stock, per trade of dollar size $Q$ ).

$$TC_i(Q) = (s_i/2) \cdot |Q| + k \cdot \sigma_i \cdot |Q|^{\{1.5\}} / ADV_i^{\{0.5\}} + \text{commission} \cdot |Q|$$

- $s_i$ : effective spread (TAQ-based where available; EDGE or AR estimator otherwise).
- $\sigma_i$ : daily volatility;  $ADV_i$ : 20-day average dollar volume.
- $k = 1$  baseline, with  $0.5\times$ ,  $2\times$  and  $3\times$  sensitivity. The  $3/2$  total-cost power corresponds to square-root impact; Almgren et al. (2005) report a concave impact exponent near  $3/5$  [verify], and the functional form is a sensitivity dimension.
- commission = 1 bp.
- Short positions pay 25 bps per year (general collateral). Stocks in the bottom institutional-ownership decile pay 5% per year, or Markit fees where licensed.
- AUM baseline \$1bn (Dec-2020 dollars, scaled by total CRSP market capitalisation), grid {\$0.1bn, \$1bn, \$10bn}.

Bounds: live institutional costs may be lower (Frazzini, Israel & Moskowitz, 2018); fund-implied costs may be higher (Patton & Weller, 2020).

**Execution assumption.** Signals are formed with information available at month-end  $t$ . Trades are assumed executed at the close of the first trading day of month  $t+1$  (one-day implementation lag), with returns accrued from that price. A same-day-close variant is reported only as a sensitivity.

## 8. Benchmarks

| Group                 | Benchmarks                                                                                                                                                                           |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Naive                 | 1/N universe; VW market; EW z-score composite; theme-EW composite; IC-weighted (rolling 60m); ICIR-weighted; inverse-volatility of signal portfolios                                 |
| Linear statistical    | Lewellen FM rolling slopes (Lewellen, 2015); ridge; lasso; elastic net; PLS (Light, Maslov & Rytchkov, 2017)                                                                         |
| Portfolio-level       | KNS shrunk SDF over signal portfolios (Kozak, Nagel & Santosh, 2020); NRRZ-style PLS factor timing (Neuhierl et al., 2024)                                                           |
| Nonlinear ML          | Random forest; LightGBM; GKX NN3 (Gu, Kelly & Xiu, 2020); IPCA expected returns (Kelly, Pruitt & Su, 2019)                                                                           |
| Economic objective    | DMNU linear PPP with TC (DeMiguel et al., 2020); JKMP Portfolio-ML using authors' public code (Jensen et al., 2026); deep PPP (Simon, Weibels & Zimmermann, 2026) if code obtainable |
| Cost mitigation       | NMV buy/hold banding applied to the EW composite (Novy-Marx & Velikov, 2016)                                                                                                         |
| Factor models (alpha) | FF5 + UMD; q5; JKP 13 themes; DNMV net-of-cost model comparison logic (Detzel, Novy-Marx & Velikov, 2023)                                                                            |

Every benchmark receives the **same** tuning budget per refit as the corresponding factorial cell, and all hyperparameter grids are published.

## 9. Validation Framework

- Walk-forward.** Initial training Jan 1972 to Dec 1989; validation Jan 1990 to Dec 1994; first test year 1995. Each January: expanding training window, rolling 5-year validation immediately before the test year, embargo gap equal to the target horizon  $h$  between training/validation and validation/test (labels overlapping the next split are purged) (Lopez de Prado, 2018). Refit annually.
- Lockbox.** Jan 2021 to Dec 2025 remains untouched until all design choices are frozen and pre-registered. It is opened once.
- Rolling-window robustness.** A 20-year rolling training window.

- **No k-fold shuffling** of time. Purged grouped cross-validation is used only inside the training window for early-stopping diagnostics.
- **Point-in-time rules.** Annual Compustat items are used  $\geq 4$  months after fiscal year-end; quarterly items after RDQ, or  $\geq 4$  months after quarter-end if RDQ is missing. Market equity is at  $t$ . Signals whose original publication year is after  $t$  are flagged, and a gated library is used in robustness (McLean & Pontiff, 2016). The CRSP-Compustat link is valid at  $t$ . Survivorship: the universe is formed at  $t$  from all securities alive at  $t$ . Delisting returns come from CRSP; missing performance-delisting returns are set to  $-30\%$  (Shumway, 1997; verify convention).

## 10. Evaluation Metrics

| Group                              | Metrics                                                                                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prediction                         | Pooled OOS $R^2$ vs zero (Gu, Kelly & Xiu, 2020); MSE; IC, rank IC, ICIR, IC t-stat (NW); top-minus-bottom decile spread (VW)                                                |
| Portfolio (gross & net)            | Annualised mean, volatility, Sharpe, Sortino, max drawdown, Calmar, skewness, kurtosis, CVaR(5%)                                                                             |
| Implementation                     | One-way monthly turnover; average holding period; cost drag (bps/month); share of costs from spread vs impact; short-leg share of returns; microcap/illiquid exposure        |
| Economic                           | Certainty-equivalent return; net alpha vs FF5+UMD, q5, JKP themes (NW t); implementable efficient frontier (Jensen et al., 2026); break-even AUM; break-even cost multiplier |
| Conversion rates (novel reporting) | Net Sharpe per unit IC; cost drag per unit turnover; transfer coefficient (correlation of scores with implemented weights) (Clarke, de Silva & Thorley, 2002)                |

## 11. Statistical Framework

1. **Mean returns and alphas.** Newey-West HAC t-statistics (12 lags) (Newey & West, 1987).
2. **Sharpe differences.** Ledoit-Wolf studentised circular block bootstrap (block lengths 4, 6, 12; 5,000 reps) (Ledoit & Wolf, 2008); Lo's serial-correlation adjustment for reporting (Lo, 2002).
3. **Forecast comparisons.** Clark-West for nested models (Clark & West, 2007); Diebold-Mariano on cross-sectional average errors (Diebold & Mariano, 1995).
4. **Data snooping.** Every configuration is logged (MLflow). Hansen SPA versus the strongest benchmark (stationary bootstrap, mean block 6, 10,000 reps) (Hansen, 2005; Politis & Romano, 1994); White Reality Check (White, 2000); Romano-Wolf StepM to identify which cells beat the benchmark (Romano & Wolf, 2005); Model Confidence Set at 10% (Hansen, Lunde & Nason, 2011).
5. **Backtest overfitting.** Deflated Sharpe ratio using the logged number of trials and their correlation (Bailey & Lopez de Prado, 2014); probability of backtest overfitting via CSCV with 16 blocks (Bailey et al., 2017); Harvey-Liu haircut Sharpe (Harvey & Liu, 2015).
6. **Factorial attribution.** Monthly net returns of the 16 cells regressed on  $\pm 1$ -coded factors and all interactions (orthogonal contrasts). Shapley decomposition of net Sharpe and CE across the four ingredients. Confidence intervals by month-block bootstrap, which preserves cross-cell dependence.
7. **Stability.** Bai-Perron structural breaks in rolling cell differences (Bai & Perron, 1998); subperiod tests.
8. **Multiple signal/theme claims.** Benjamini-Hochberg FDR (Benjamini & Hochberg, 1995);  $t > 3$  hurdle for any new-signal-type claim (Harvey, Liu & Zhu, 2016).
9. **Non-standard errors.** Distribution of main-effect estimates across a pre-declared multiverse (window type, universe, horizon, target transform, cost estimator, risk model) (Menkveld et al., 2024; Chen, Hanauer & Kalsbach, 2024).

## 12. Robustness Framework

Subperiods (1995-2000, 2001-2008, 2009-2019, 2020-2025; pre/post decimalisation). Universes (all stocks incl. microcaps; ex-microcaps (main); NYSE top 500). Weighting and constraints (long-only; no industry constraint). Horizons (1, 3, 6, 12m) and quarterly rebalancing. Cost multipliers 0.5-3× and alternative spread estimators. AUM grid. Risk models (structural, PCA, LW shrinkage). Rolling vs expanding windows. Publication-gated signal library. Placebo states. NN seeds ( $\geq 10$ ) and  $\pm 1$  grid-step perturbations. International out-of-sample (JKP developed ex-US). Alternative uncertainty estimators (bootstrap SE, MC dropout, conformal).

## 13. Economic Interpretation Plan

- **Economic feature importance.** Change in net utility when a theme is removed and the model refit (Jensen et al., 2026). TreeSHAP and deep SHAP by theme for P-cells (Lundberg & Lee, 2017). ALE plots, given correlated signals (Apley & Zhu, 2020).
- **Interaction analysis.** Friedman H-statistics for the top theme pairs; comparison of gross vs net importance.
- **State dependence.** Implied signal-theme weights (cross-sectional regression of implemented weights on theme scores) regressed on  $z_t$ , with tests of  $H_5$ .
- **Portfolio attribution.** Returns decomposed into theme exposures, industry, market and residual; cost attribution by theme.
- **Limits to arbitrage.** Portfolio tilts on idiosyncratic volatility, illiquidity, size and distress by cell ( $H_6$ ) (Avramov, Cheng & Metzker, 2023).
- **Alignment.** Ex-ante vs ex-post risk bias statistics by cell (Ceria, Saxena & Stubbs, 2012).

## 14. Contribution Statement (draft)

*Existing literature shows that machine-learning combinations of firm characteristics improve return prediction (Gu, Kelly & Xiu, 2020), that the resulting portfolios are fragile to economic restrictions and trading costs (Avramov, Cheng & Metzker, 2023; Blitz et al., 2023), and that learning portfolio weights under a cost-aware objective substantially improves the implementable efficient frontier (Jensen et al., 2026; DeMiguel et al., 2020; Simon, Weibels & Zimmermann, 2026).*

*However, these studies compare bundled pipelines. They largely assume that the sources of ML's value (flexible functional form, time-varying signal efficacy, the training objective and forecast uncertainty) can be assessed one at a time, on different data, costs and evaluation criteria.*

*We propose a controlled factorial design that switches each ingredient on and off within one signal library, universe, optimiser, cost model and inference framework, and we attribute net-of-cost performance to the ingredients and their interactions with data-snooping-robust tests.*

*This allows us to test whether nonlinearity creates value beyond its trading costs, whether state-dependent signal weights help only when costs are internalised, and whether uncertainty shrinkage works through turnover.*

*We demonstrate [results to be determined] across US equities from 1995 to 2025, including an untouched 2021-2025 holdout, three AUM levels, cost-model perturbations and developed international markets.*

*The results imply [to be determined; stated only after the lockbox is opened].*

**What we explicitly do not claim.** We are not the first to use ML for signal combination, the first to include transaction costs, the first to condition on market states, or the first to quantify design-choice dispersion (see the closest-prior-art tracker).

## 15. Comparison Tables Against the Closest Papers

### 15.1 Ten closest papers

| Paper                                  | What it does                                           | What we use                                                 | What we improve                                                              | Why ours is different                                          |
|----------------------------------------|--------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|
| Jensen et al. (2026)                   | Cost-aware ML weight learning; implementable frontier  | Frontier evaluation; Portfolio-ML as benchmark; public code | Isolate the objective effect from nonlinearity, conditioning and uncertainty | Attribution and interaction tests, not a single pipeline       |
| Simon, Weibels & Zimmermann (2026)     | NN parametric portfolio policies vs linear             | Nonlinear economic-loss cell                                | Common cost model/optimiser; conditioning; snooping control                  | Decomposition and mechanisms                                   |
| DeMiguel et al. (2020)                 | Linear PPP with TC; netting                            | Linear economic-loss cell                                   | Nonlinear and conditional extensions                                         | Measures netting benefits of nonlinear/conditional combination |
| DeMiguel, Martin-Utrera & Uppal (2024) | Conditional multifactor weights net of costs           | Conditioning-with-costs logic                               | Stock-level, many states, nonlinear                                          | Tests the conditioning × cost interaction at scale             |
| Gu, Kelly & Xiu (2020)                 | ML prediction horse race                               | Model classes and OOS R <sup>2</sup> design                 | Net-of-cost evaluation in a common optimiser                                 | Economic rather than statistical attribution                   |
| Avramov, Cheng & Metzker (2023)        | ML under economic restrictions                         | Universe restrictions; limits-to-arbitrage tests            | Constructive design                                                          | Explains which ingredient causes fragility                     |
| Neuhierl et al. (2024)                 | High-dimensional factor timing                         | State variables; PLS timing benchmark                       | Stock-level implementation with netting and costs                            | Timing information inside cost-aware combination               |
| Chen, Hanauer & Kalsbach (2024)        | ML design choices; non-standard errors                 | Multiverse reporting                                        | Economic ingredients; net of costs; common optimiser                         | Attribution of net value, not dispersion                       |
| Liao et al. (2025)                     | NN forecast uncertainty; uncertainty-averse portfolios | Bootstrap SE variant                                        | Cost interaction; turnover channel                                           | Uncertainty as one factor in the factorial design              |
| Blitz et al. (2023)                    | Horizon and net ML alpha                               | Horizon grid                                                | Horizon chosen in validation and counted as trials                           | Horizon as a controlled design factor                          |

### 15.2 Five strongest methodological precedents

| Paper                  | What it does        | What we use                                     | What we improve            | Why ours is different |
|------------------------|---------------------|-------------------------------------------------|----------------------------|-----------------------|
| Hansen (2005)          | SPA test            | Snooping control over cells                     | Applied to factorial cells | Cell-level inference  |
| Romano & Wolf (2005)   | StepM               | Identify cells beating benchmark                | -                          | -                     |
| Bailey et al. (2017)   | PBO via CSCV        | Overfitting probability of chosen configuration | Report for full trial log  | Transparency          |
| Menkveld et al. (2024) | Non-standard errors | Multiverse design                               | Economic ingredients       | Attribution           |
| Ledoit & Wolf (2008)   | Robust Sharpe tests | Pairwise Sharpe inference                       | Gross vs net               | Conversion rates      |

### 15.3 Five strongest portfolio-construction precedents

| Paper                                   | What it does                              | What we use                        | What we improve                    | Why ours is different  |
|-----------------------------------------|-------------------------------------------|------------------------------------|------------------------------------|------------------------|
| Garleanu & Pedersen (2013)              | Dynamic trading with costs; aim portfolio | Partial adjustment logic           | Learned signal weights             | Empirical ML setting   |
| Collin-Dufresne, Daniel & Saglam (2020) | Liquidity-regime allocation with costs    | H5 trading-speed prediction        | Empirical test at stock level      | Theory test            |
| Brandt, Santa-Clara & Valkanov (2009)   | Parametric portfolio policies             | Economic-loss parameterisation     | Nonlinear, conditional, cost-aware | Nested cells           |
| Ledoit & Wolf (2017)                    | Nonlinear shrinkage covariance            | Robustness risk model              | -                                  | -                      |
| Novy-Marx & Velikov (2016)              | Anomaly costs and mitigation              | Banding benchmark; spread measures | Optimised mitigation               | Optimiser vs heuristic |

### 15.4 Design v2: what changed after the 2025-2026 frontier search (September 2026)

A targeted search of what these journals have accepted or circulated in 2025-2026 produced four changes. Each is a **level on an existing axis** or a **credibility instrument**; nothing was added as decoration, because every extra arm weakens the deflated Sharpe ratio of the eventual winner.

| Change                      | What it is                                                                                                                                                      | Why it is now necessary                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form ladder A2-A4           | Ridgeless random Fourier features on a complexity grid ( $c = P/T$ ), a sparse basis-pursuit arm on the same feature map, and a cross-sectional attention block | The complexity debate moved on: Afsharhajari & Li (2026) shows complexity pays through sparse discovery while dense ridgeless plateaus, and Kelly et al. (2025) shows it pays only when the factor spectrum is diffuse. Attention is the current state of the art for cross-asset combination (Kelly et al., 2025; Lai, 2025) |
| Uncertainty ladder D2-D3    | Time-ordered split conformal intervals with adaptive recalibration, feeding an ellipsoidal or Wasserstein robust objective                                      | Ensemble dispersion has no coverage guarantee; conformal does (Kato, 2024), and connecting it to the decision is the economically interesting claim                                                                                                                                                                           |
| Economic-objective controls | Output clipping, partial adjustment and a reported prediction-inflation diagnostic                                                                              | Wang & Hasuikie (2026) shows decision-focused learning inflates signals and turnover, which is exactly the pathology our objective axis tests                                                                                                                                                                                 |
| Credibility layer           | Zero-predictability synthetic null, microstructure placebo, multiplicity-adjusted inflation gap, spectrum diagnostic, limits-to-learning caveat                 | [verify authors] (2026) shows adaptive specification search produces significant backtests under a no-predictability null; Chen, Kelly & Malamud (2025) shows empirical fit understates true predictability                                                                                                                   |

**New closest prior art.** Zhang (2025) (October 2025) performs friction-aware, regime-conditioned policy optimisation with costs inside the reward and a regime x cost evaluation grid. It is the nearest neighbour to gap G2 and must be cited in the introduction. The differentiation is structural: it proposes a single reinforcement-learning method in a small asset space, whereas this paper identifies *which ingredient* produces net-of-cost value in a ~150-signal stock cross-section under a nested factorial design with data-snooping-robust inference.

**Deliberately excluded**, with reasons recorded so referees see the choice was made rather than missed: deep reinforcement-learning agents (occupied by Zhang (2025); replication risk), graph neural networks on licensed relationship data (point-in-time integrity), LLM-generated signals (training data covers our sample, so look-ahead is unresolvable before a strict post-cutoff window), and diffusion models for return distributions (no finance precedent; adds trials without touching an axis).

## 15.5 Five strongest ML-finance precedents

| Paper                             | What it does                       | What we use                   | What we improve                          | Why ours is different      |
|-----------------------------------|------------------------------------|-------------------------------|------------------------------------------|----------------------------|
| Gu, Kelly & Xiu (2020)            | ML prediction benchmark            | NN3/GBDT designs              | Net evaluation                           | Attribution                |
| Chen, Pelger & Zhu (2024)         | State-dependent deep SDF           | Macro-state conditioning idea | Costs; interpretable fixed states        | Implementable conditioning |
| Kelly, Pruitt & Su (2019)         | IPCA                               | Benchmark expected returns    | -                                        | -                          |
| Azevedo, Hoegner & Velikov (2023) | ML strategies net of costs         | Net evaluation conventions    | Cost-aware learning                      | Attribution                |
| Cong et al. (2021)                | RL/transformer direct construction | Direct-objective idea         | Controlled comparison under strict costs | Replicability focus        |

# Part III. Experimental Framework and Implementation Specification

This document is written to be handed directly to a quant developer. Parameter values are the **proposed baseline**. Any change after the lockbox specification freeze (end of Phase 7) must be logged as a deviation.

## 1. End-to-End Research Pipeline

| #  | Stage                       | What happens                                                                                           | Key controls                                              | Output artefact        |
|----|-----------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------|
| 1  | Data                        | Pull CRSP, Compustat, CCM, TAQ spreads, factors, states                                                | Versioned raw extracts with pull dates                    | data/raw/* + manifest  |
| 2  | Point-in-time engineering   | Lag accounting data, link identifiers at t, apply delisting returns, build month-end panels            | Automated look-ahead unit tests                           | panel_pit.parquet      |
| 3  | Feature / signal generation | Compute ~150 JKP/OpenAP signals; publication-year flags                                                | Replication tests vs published t-stats                    | signals.parquet        |
| 4  | Signal cleaning             | Remove signals with <50% average coverage; handle outliers via ranks                                   | Coverage report by year                                   | signal_qc.csv          |
| 5  | Normalisation               | Cross-sectional rank to [-0.5, 0.5]; missing -> 0 + theme missing indicators                           | Only information at t                                     | X_t arrays             |
| 6  | Signal quality analysis     | IC, rank IC, ICIR by horizon; IC half-life; turnover of signal portfolios                              | NW t-stats                                                | E02 report             |
| 7  | Redundancy analysis         | Rank-correlation and return-correlation clustering; effective number of signals                        | Bootstrap stability                                       | E03 report             |
| 8  | Train/validation/test       | Expanding train from 1972, 5y validation, embargo = h, annual refit, 1995-2020 test, 2021-2025 lockbox | Split calendar file checked into repo                     | splits.yaml            |
| 9  | ML alpha combination        | 16 factorial cells + benchmarks                                                                        | Identical tuning budgets; all trials logged               | Model registry         |
| 10 | Expected return / ranking   | P-cells -> standardized scores -> IC-scaled alpha; E-cells -> weight proposals                         | Common scaling rule                                       | alpha_t, w_prop_t      |
| 11 | Portfolio construction      | TC-aware MVO with constraints; E-cell projection                                                       | Solver status logs                                        | weights_t              |
| 12 | Risk controls               | Dollar/beta/industry neutrality; position, trade and participation limits                              | Constraint-violation checks                               | Risk report            |
| 13 | Transaction costs           | Spread + power-law impact + commission + borrow                                                        | Cost multipliers; AUM grid                                | Cost ledger            |
| 14 | Execution assumptions       | Trade at close of first trading day of t+1; returns from execution price                               | One-day lag enforced in code                              | Execution log          |
| 15 | Backtesting                 | Monthly P&L gross/net; drifted weights                                                                 | Reconciliation: $\Sigma$ contributions = portfolio return | returns_cells.parquet  |
| 16 | Statistical tests           | NW, LW2008, CW, DM, SPA, StepM, MCS, DSR, PBO                                                          | Trial log feeds DSR/PBO                                   | Inference tables       |
| 17 | Robustness                  | Subperiods, universes, costs, AUM, horizons, risk models, placebo, gating, seeds, international        | Pre-declared multiverse                                   | Robustness appendix    |
| 18 | Economic interpretation     | Economic feature importance, SHAP/ALE, implied state-dependent weights, limits-to-arbitrage tilts      | Tie each result to hypothesis H1-H6                       | Interpretation section |

## 2. Specification Summary (configuration)

```

universe:
  source: CRSP
  share_codes: [10, 11]
  exchanges: [NYSE, AMEX, NASDAQ]
  main_filter: {min_price: 5, exclude_below_nyse_size_pct: 20}
  robustness_universes: [all_stocks, nyse_top500]
frequency: {signals: monthly, rebalancing: monthly, risk_and_costs: daily}
periods:
  initial_train: [1972-01, 1989-12]
  initial_validation: [1990-01, 1994-12]
  walk_forward_test: [1995-01, 2020-12]
  lockbox: [2021-01, 2025-12]
  refit: annual_january
  train_window: expanding # robustness: rolling_240m
  validation_window: 60m_rolling
  embargo_months: horizon_h
signals:
  library: JKP_characteristics # cross-check: OpenAP
  themes: 13
  normalisation: cross_sectional_rank[-0.5,0.5]
  missing: zero_plus_theme_indicators
  publication_gating: robustness_only
states: [MKTVOL, BEAR, ILLIQ, SENT, CREDIT, TERM, INFL, DRATE, FMOM_theme, DISP]
targets: {prediction_horizons: [1, 3, 6, 12], choose_in_validation: true}
factorial:
  form: [linear, nonlinear]
  conditioning: [static, state]
  objective: [prediction, economic]
  uncertainty: [none, ensemble_shrinkage]
models:
  ridge: {lambda_grid: logspace(-4, 4, 17)}
  lightgbm: {num_leaves: [15, 31, 63], learning_rate: [0.01, 0.05], min_data_in_leaf: [500, 2000],
    feature_fraction: [0.5, 0.8], bagging_fraction: 0.8, lambda_l2: [0, 10],
    max_trees: 2000, early_stopping: validation}
  nn3: {layers: [32, 16, 8], activation: relu, batch_norm: true, optimizer: adam, lr: 0.001,
    l1: [1e-5, 1e-4, 1e-3], early_stopping_patience: 5, ensemble_seeds: 10}
  moe: {experts: 3, gate_inputs: states_only}
  economic_policy: {net: nn3, bptt_months: 12, gamma: calibrate_to_10pct_vol, huber_delta: 1e-4}
  uncertainty: {members: 10, kappa_grid: [0, 0.5, 1, 2, 4]}
portfolio:
  objective: mean_variance_minus_tc
  constraints: {dollar_neutral: true, gross_max: 2.0, beta_abs_max: 0.05,
    industry_ff49_abs_max: 0.02, weight_abs_max: 0.01, adv_participation_max: 0.05}
  long_only_variant: {tracking_error_max: 0.04}
  solver: [MOSEK, CLARABEL]
risk_model:
  type: structural
  factors: [market_beta, jkp_13_themes, ff49_industries]
  factor_cov: {ewma_halflife_days: 126, newey_west: true}
  specific: {ewma_halflife_days: 126, bayesian_shrink: size_deciles}
  robustness: [pca_20, ledoit_wolf_top500]
costs:
  spread: {primary: taq_effective_spread, fallback: [edge_ohlc, abdi_ranaldo]}
  impact: {form: "k * sigma_daily * |Q|^{1.5} / ADV^{0.5}", k: 1.0, sensitivity: [0.5, 2.0, 3.0]}
  commission_bps: 1
  borrow: {general_collateral_bps_pa: 25, hard_to_borrow_pa: 0.05, htb_proxy: bottom_decile_inst_ownership}
  aum_usd_2020: [1.0e8, 1.0e9, 1.0e10]
  aum_scaling: total_crsp_market_cap
execution: {lag_days: 1, price: close}
inference:
  nw_lags: 12
  lw2008_blocks: [4, 6, 12]
  spa: {bootstrap: stationary, mean_block: 6, reps: 10000}
  mcs_alpha: 0.10
  pbo_cscv_blocks: 16
  bootstrap_reps: 5000
logging: {tracker: mlflow, log_every_trial: true}

```

## 3. Experiment Register

The experiment register (IDs E00-E64: hypothesis, model, dataset, benchmark, metrics, statistical test, expected outcome, status) is in Workbook 4. Summary by block:

| Block          | Experiments | Purpose                                                         |
|----------------|-------------|-----------------------------------------------------------------|
| Data integrity | E00         | PIT audit and factor replication                                |
| Signal library | E01-E03     | Replication, quality/decay, redundancy                          |
| Baselines      | E10-E14     | Naive, linear, portfolio-level, published ML, linear cost-aware |

| Block             | Experiments | Purpose                                                                                     |
|-------------------|-------------|---------------------------------------------------------------------------------------------|
| Factorial core    | E20-E29     | 16 cells + decomposition                                                                    |
| Portfolio & costs | E30-E34     | Frontier, cost sensitivity, capacity, constraints, heuristics                               |
| Inference         | E40-E43     | SPA/StepM/MCS, DSR/PBO, alphas, multiverse                                                  |
| Robustness        | E50-E56     | Subperiods, universes, horizons, placebo, gating, international, seeds                      |
| Interpretation    | E60-E64     | Economic importance, state-dependent weights, limits to arbitrage, trading speed, alignment |

## 4. Bias Controls Matrix

| Bias                             | Where it enters                                                  | Control in this design                                                                     |
|----------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Look-ahead                       | Accounting lags; regime labels; standardisation; hyperparameters | PIT rules; filtered states only; cross-sectional ranks; validation-only tuning; unit tests |
| Survivorship                     | Universe construction                                            | Universe at t from all live securities; delisting returns                                  |
| Selection bias                   | Choosing signals, periods, universes after seeing results        | Signal library fixed from JKP/OpenAP before modelling; pre-registration; lockbox           |
| Data snooping / multiple testing | Many cells and configurations                                    | Full trial logging; SPA, StepM, MCS; DSR; PBO; haircuts                                    |
| Overfitting                      | Complex models                                                   | Early stopping; ensembles; validation windows; PBO                                         |
| Leakage                          | Overlapping multi-month labels                                   | Purging and embargo equal to horizon                                                       |
| Rebalancing assumptions          | Same-close trading                                               | One-day execution lag; drifted weights                                                     |
| Delisting bias                   | Missing delisting returns                                        | CRSP dlret with convention for missing values                                              |
| Corporate actions                | Price/share adjustments                                          | CRSP adjustment factors; total returns                                                     |
| Point-in-time information        | Restatements; macro revisions                                    | Lags; Snapshot robustness; ALFRED or market-based states                                   |

## 5. Validation Approaches: Which Are Appropriate

| Approach                               | Appropriate here?                                   | Reason                                                       |
|----------------------------------------|-----------------------------------------------------|--------------------------------------------------------------|
| Walk-forward with expanding windows    | <b>Yes (main)</b>                                   | Mimics real-time research; maximises data for complex models |
| Rolling windows                        | Yes (robustness)                                    | Addresses non-stationarity (Capponi et al., 2025)            |
| Nested cross-validation (time-ordered) | Yes, for hyperparameters within the training window | Outer loop = walk-forward; inner loop = validation block     |
| Purged k-fold with embargo             | Yes, inside training for early-stopping diagnostics | Required with overlapping labels (Lopez de Prado, 2018)      |
| Standard shuffled k-fold               | <b>No</b>                                           | Leaks temporal and cross-sectional dependence                |
| Combinatorial purged CV (CSCV)         | Yes, for PBO estimation only                        | (Bailey et al., 2017)                                        |
| Single lockbox holdout                 | <b>Yes (final)</b>                                  | Only credible defence against iterative snooping             |

## 6. Reviewer-Proof Checklist (summary)

See Reviewer\_Proof\_Checklist.md for the full list with owners and evidence locations.

## 7. Implementation Roadmap (summary)

See Research\_Roadmap.md.

## Part IV. Reviewer-Proof Checklist

Tick every item before submission, and record where the evidence lives (table, figure, appendix or code path). Items are derived from the published critiques and standards cited in this package (Harvey, Liu & Zhu, 2016; Harvey, 2017; Arnott, Harvey & Markowitz, 2019; Avramov, Cheng & Metzker, 2023; Chen, Hanauer & Kalsbach, 2024; Menkveld et al., 2024; Jensen et al., 2026) and from journal code/data policies (JF, RFS, MS, verified).

### A. Contribution and positioning

- ☐ The research question is stated in one sentence as an economic question, not a model description.
- ☐ The contribution statement names the closest papers (JKMP2026, SWZ2023, DMNU2020, DMU2024, GKX2020, ACM2023, CHK2024, LMNS2025) and says what is new relative to each.
- ☐ The closest-prior-art tracker has been re-run within 2 weeks of submission (SSRN, NBER, arXiv, conference programmes, forward citations).
- ☐ No "first to" claim appears unless the tracker supports it.
- ☐ The hypotheses were pre-registered (dated document or OSF/AEA registry) before the lockbox was opened.
- ☐ Null or negative results are reported, with power discussion.

### B. Data and point-in-time integrity

- ☐ All data sources, pull dates and versions are documented.
- ☐ Accounting variables are lagged by availability ( $\geq 4$  months annual; RDQ quarterly) and unit tests prove it.
- ☐ Universe formed at  $t$  from all live securities (no survivorship).
- ☐ CRSP delisting returns are applied; the convention for missing values is documented.
- ☐ Corporate actions are handled via CRSP adjustment factors and total returns.
- ☐ Microcaps are excluded from the main universe; results including them are in robustness.
- ☐ Macro/state variables use real-time (vintage) data or market-based measures; no smoothed regime probabilities.
- ☐ Rebuilt standard factors correlate  $> 0.95$  with the Ken French library.
- ☐ Signal replication statistics versus Chen-Zimmermann/JKP are reported.
- ☐ Missing-data handling is described, with robustness to imputation methods (Bryzgalova et al., 2025; Freyberger et al., 2025).

### C. Validation and leakage

- ☐ Walk-forward splits, refit schedule, embargo and purging are documented in a split calendar.
- ☐ No hyperparameter is tuned on test data; the validation windows strictly precede test years.
- ☐ Lockbox period (2021-2025) opened once, after the specification freeze; the freeze date is recorded.
- ☐ Every configuration tried is logged (count reported in the paper).
- ☐ Label overlap for multi-month horizons is purged.
- ☐ NN results report dispersion across  $\geq 10$  seeds.

### D. Benchmarks

- ☐ Naive composites (EW, theme-EW, IC-weighted, ICIR-weighted, inverse-vol) are included.
- ☐ Linear baselines (Lewellen FM, ridge, ENet, PLS) are included.
- ☐ Strongest published implementable benchmarks are included (JKMP Portfolio-ML from authors' code; DMNU linear PPP).
- ☐ Benchmarks receive the same tuning budget and cost model as proposed cells.
- ☐  $1/N$  and market portfolios are reported (DeMiguel, Garlappi & Uppal, 2009).
- ☐ Heuristic cost mitigation (banding) is compared (Novy-Marx & Velikov, 2016).

## E. Portfolio construction, costs and capacity

- ☐ Net-of-cost results are primary; gross results are secondary.
- ☐ Cost model components (spread, impact, commission, borrow) and calibration sources are stated.
- ☐ Sensitivity to cost multipliers ( $0.5\times-3\times$ ) and alternative spread estimators is shown.
- ☐ AUM grid and break-even AUM (capacity) are reported.
- ☐ Turnover, holding period and cost drag are reported for every strategy.
- ☐ Constraint set (neutralities, position/trade/participation limits) is identical across cells.
- ☐ Execution lag ( $\geq 1$  trading day) is enforced.
- ☐ Short-leg contribution and borrow-cost treatment are disclosed.
- ☐ Long-only variant reported.

## F. Statistical inference

- ☐ HAC (Newey-West) t-stats for means and alphas.
- ☐ Sharpe-ratio differences tested with Ledoit-Wolf (2008) bootstrap.
- ☐ Nested forecast comparisons with Clark-West; others with Diebold-Mariano.
- ☐ SPA and Reality Check versus the strongest benchmark; StepM for which cells win; MCS reported.
- ☐ Deflated Sharpe ratio uses the logged trial count; PBO via CSCV reported.
- ☐ Factorial effects and Shapley attributions have bootstrap confidence intervals.
- ☐ Alphas versus FF5+UMD, q5 and JKP themes (net of costs).
- ☐ Multiple theme/signal claims are FDR-controlled.
- ☐ Non-standard errors across the pre-declared multiverse are reported.

## G. Robustness

- ☐ Subperiods, including pre/post decimalisation and post-2008.
- ☐ Universes: all stocks, ex-micro, top-500.
- ☐ Horizons and rebalancing frequency.
- ☐ Risk-model alternatives.
- ☐ Rolling vs expanding windows.
- ☐ Placebo (shuffled) states.
- ☐ Publication-gated signal library.
- ☐ International (developed ex-US) evidence or a stated reason for exclusion.
- ☐ Hyperparameter perturbations.

## H. Economic interpretation

- ☐ Economic feature importance (net utility loss from dropping themes).
- ☐ SHAP/ALE interpretation for prediction cells, with correlated-feature caveats.
- ☐ Implied state-dependent weights are linked to DM2016, SYY2012 and CDS2020 predictions.
- ☐ Limits-to-arbitrage tilts (IVOL, illiquidity, size, distress) by cell.
- ☐ A clear answer to "why does it work (or fail)?" appears in the introduction.

## I. Reproducibility and presentation

- ☐ Replication package: code, environment file, README, pseudo-data for licensed inputs (JF/RFS/MS policies).
- ☐ Figures readable in grayscale; tables have units and sample periods.
- ☐ The abstract states net-of-cost results and the holdout period.
- ☐ Internet appendix contains all robustness tables referenced.

- **Limitations** section covers cost-model uncertainty, US focus, and data licences.

## Part V. Research Roadmap (staged implementation plan)

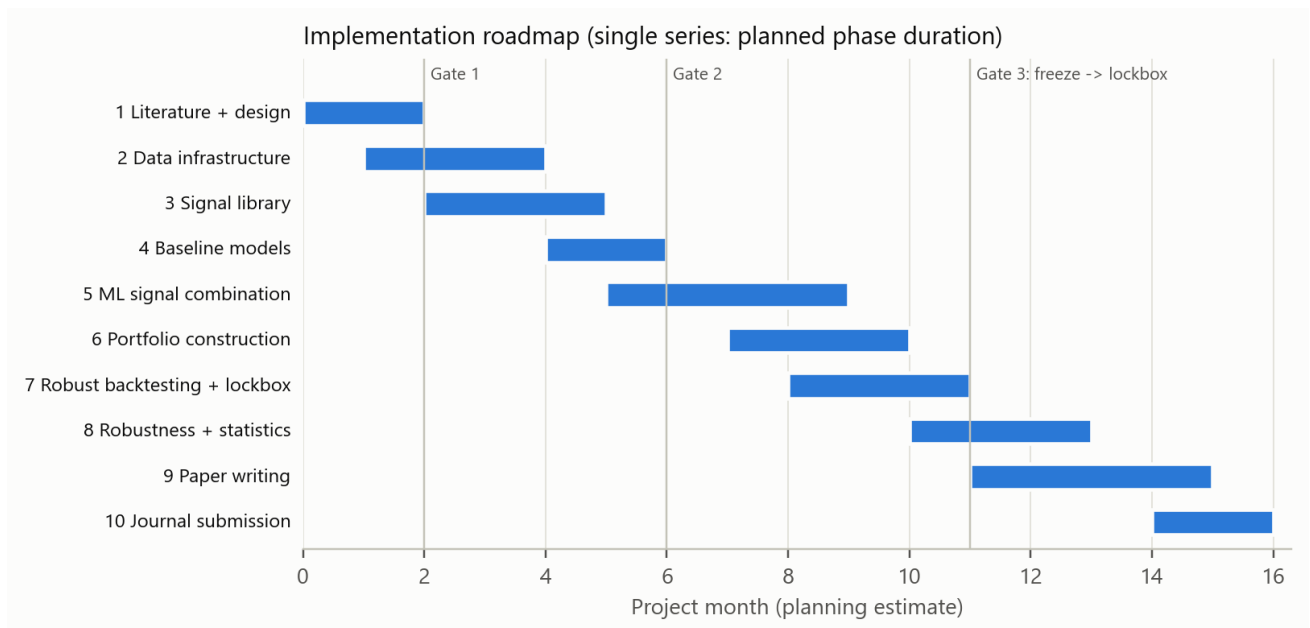


Figure. Implementation roadmap with go/no-go gates (planning estimates).

Durations are planning estimates for a small team (1-2 researchers plus part-time developer support), in months from project start. They are not commitments. Journal review times vary widely and are not included in the 16-month build.

| Phase                           | Months | Tasks                                                                                                                               | Dependencies     | Deliverables                                                 | Key risks                                   | Acceptance criteria                                                         |
|---------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------|
| 1. Literature + research design | 1-2    | Finalise closest-prior-art search; pre-register hypotheses H1-H6; freeze factorial design and benchmark list; confirm data licences | This package     | Pre-registration document; updated tracker; design spec v1.0 | Newly published overlapping paper           | Tracker re-run; no overlap triggering the re-scope rule; co-author sign-off |
| 2. Data infrastructure          | 2-4    | WRDS extracts (CRSP, Compustat, CCM, TAQ spreads); PIT engineering; delisting handling; state variables; unit tests                 | Phase 1 licences | Versioned PIT panel; data manifest; test suite               | Licence gaps (TAQ, Markit); PIT errors      | E00 passes: factor correlations > 0.95; zero look-ahead test failures       |
| 3. Alpha signal library         | 3-5    | Build ~150 signals (JKP code / OpenAP); publication flags; quality, decay, redundancy diagnostics                                   | Phase 2          | Signal panel; E01-E03 reports                                | Signal definition mismatches                | Replication slope/ $R^2$ close to CZ2022 benchmarks; coverage report        |
| 4. Baseline models              | 5-6    | Naive, linear, KNS SDF, published ML, DMNU PPP, JKMP Portfolio-ML replication                                                       | Phase 3          | E10-E14 results; benchmark registry                          | JKMP code compute burden                    | Directional replication of published benchmarks documented                  |
| 5. ML signal combination        | 6-9    | Implement 16 factorial cells; uncertainty ensembles; trial logging (MLflow)                                                         | Phases 3-4       | Model registry; E20-E29 development results                  | Compute; unstable NN economic-loss training | All cells run end-to-end on the 1995-2020 walk-forward; seeds logged        |
| 6. Portfolio construction       | 8-10   | Risk model; TC-aware optimiser; constraints; cost model; AUM grid; E-cell projection                                                | Phases 2, 5      | Optimiser module; E30-E34                                    | Solver failures; cost calibration disputes  | Constraint-violation rate < 0.1% of months; cost sensitivity grid complete  |
| 7. Robust backtesting           | 9-11   | Execution lag; reconciliation; multiverse runs; <b>specification freeze; then open the lockbox once</b>                             | Phases 5-6       | Frozen spec (dated); lockbox results                         | Temptation to iterate after lockbox         | Freeze date precedes lockbox run in logs                                    |

| Phase                               | Months                  | Tasks                                                                                                                                       | Dependencies | Deliverables                                                  | Key risks                        | Acceptance criteria                                                            |
|-------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------|
| 8. Robustness + statistical testing | 11-13                   | SPA/StepM/MCS, DSR, PBO, factorial & Shapley attribution, E40-E56; interpretation E60-E64                                                   | Phase 7      | Inference tables; robustness appendix; interpretation figures | Weak power for interactions      | All pre-registered tests reported; checklist sections C-H complete             |
| 9. Paper writing                    | 12-15                   | Draft introduction and contribution; results; internet appendix; replication package; internal review; SSRN posting; conference submissions | Phase 8      | Working paper; replication package; slides                    | Over-claiming; unclear narrative | Reviewer-proof checklist 100% ticked; two external readers' comments addressed |
| 10. Journal submission              | 15-16 (+ review cycles) | Final prior-art check; journal choice (RFS stretch vs JFQA/MS); cover letter; respond to referees                                           | Phase 9      | Submission; response letters                                  | Desk rejection; long review      | Submission package meets journal policy; tracker re-run within 2 weeks         |

## Critical path

Data licences -> PIT panel -> signal library -> factorial cells -> optimiser/costs -> specification freeze -> lockbox -> inference -> writing.

## Go / no-go gates

- End of Phase 1.** If a newly found paper performs the same factorial attribution, re-scope to G2/G3 (conditioning  $\times$  cost interaction and mechanisms) or to the uncertainty and international extensions.
- End of Phase 4.** If JKMP Portfolio-ML cannot be run, substitute DMNU PPP and a deep PPP as the economic-objective benchmarks, and disclose this.
- End of Phase 7.** Whatever the lockbox shows, the paper proceeds. The narrative adapts; the design does not.

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